

BJT Bias Circuits and the Three Basic Amplifier Configurations

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1 How this document fits with the previous notes

The previous notes established the Q-point, operating regions, load lines, and the hybrid- π small-signal model. We now answer two practical questions:

1. How do common bias circuits establish the Q-point?
2. Once biased, what gain and input/output resistances do different amplifier connections provide?

All circuits below use an NPN transistor operating in its *forward-active region*. We analyze DC bias first, then replace the transistor with its small-signal model.

At room temperature, we use

$$g_m = \frac{I_{CQ}}{V_T}, \quad r_\pi = \frac{\beta}{g_m}, \quad r_e \equiv \frac{1}{g_m} \approx \frac{V_T}{I_{CQ}}, \quad V_T \approx 26 \text{ mV}. \quad (1)$$

Here r_e is the transistor's *intrinsic small-signal* resistance; R_E denotes an *external* emitter resistor.

Analysis convention. Unless stated otherwise, voltage gain means the gain from the voltage *at the amplifier input terminal* to its output. Source resistance can reduce the voltage that actually reaches that terminal; we account for it separately.

2 Fixed bias

2.1 Circuit and DC equations

In the simplest fixed-bias CE circuit, a resistor R_B connects V_{CC} to the base, R_C connects V_{CC} to the collector, and the emitter is grounded. For DC analysis, any input or output coupling capacitors are open circuits.

Using $V_{BEQ} \approx 0.7 \text{ V}$, the base loop gives

$$I_{BQ} \approx \frac{V_{CC} - V_{BEQ}}{R_B}. \quad (2)$$

If the resulting operating point is forward-active,

$$I_{CQ} \approx \beta I_{BQ}, \quad V_{CEQ} = V_{CC} - I_{CQ} R_C. \quad (3)$$

The equations must be checked after calculation. If they predict a negative V_{CE} , or a voltage below the approximate saturation voltage, the assumption of forward-active operation was invalid.

2.2 Numerical example

Take

$$V_{CC} = 12 \text{ V}, \quad R_B = 188 \text{ k}\Omega, \quad R_C = 1 \text{ k}\Omega, \quad \beta = 100.$$

Then

$$\begin{aligned} I_{BQ} &\approx \frac{12 - 0.7}{188 \text{ k}\Omega} \approx 60.1 \text{ }\mu\text{A}, \\ I_{CQ} &\approx 100(60.1 \text{ }\mu\text{A}) \approx 6.01 \text{ mA}, \\ V_{CEQ} &\approx 12 - (6.01 \text{ mA})(1 \text{ k}\Omega) \approx 5.99 \text{ V}. \end{aligned}$$

The Q-point is near the midpoint of the collector's available DC voltage range.

2.3 Why fixed bias is poorly stabilized

Equation (3) shows that I_{CQ} is approximately proportional to β . If we replace the transistor with another whose β is 150, while retaining the same circuit, the approximate collector current becomes

$$I_{CQ} \approx 150(60.1 \text{ }\mu\text{A}) \approx 9.02 \text{ mA}.$$

The new collector voltage is only about 2.98 V. Thus transistor variation moves the Q-point substantially.

Temperature-induced changes can also move the current. This simple fixed-bias circuit provides no emitter-resistor feedback to oppose that movement.

Intuition. Fixed bias approximately sets *base current*. Collector current is then whatever that base current produces for the particular transistor's β . Since β varies, so does the Q-point.

3 Voltage-divider bias with an emitter resistor

3.1 Circuit idea

Now connect R_1 from V_{CC} to the base, R_2 from base to ground, and R_E from emitter to ground. The two base resistors form a voltage divider. The emitter resistor supplies negative feedback.

We can replace the base divider by its Thevenin equivalent:

$$\boxed{V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2}, \quad R_{TH} = R_1 \parallel R_2.} \quad (4)$$

The equivalent base circuit is a voltage source V_{TH} in series with R_{TH} .

3.2 Deriving the Q-point

Apply Kirchhoff's voltage law through the Thevenin source, base-emitter junction, and emitter resistor:

$$V_{TH} = I_B R_{TH} + V_{BE} + I_E R_E.$$

Using $I_E = (\beta + 1)I_B$, we obtain

$$\boxed{I_{BQ} = \frac{V_{TH} - V_{BEQ}}{R_{TH} + (\beta + 1)R_E}}. \quad (5)$$

The remaining quantities follow from

$$I_{CQ} = \beta I_{BQ}, \quad I_{EQ} = (\beta + 1)I_{BQ}, \quad (6)$$

$$V_{EQ} = I_{EQ}R_E, \quad V_{CQ} = V_{CC} - I_{CQ}R_C, \quad (7)$$

$$\boxed{V_{CEQ}} = V_{CQ} - V_{EQ}. \quad (8)$$

Again, check that the calculated transistor is forward-active. In particular, its collector should be sufficiently above its emitter and the base-collector junction should be reverse biased.

3.3 Numerical example

Choose

$$V_{CC} = 12 \text{ V}, \quad R_1 = 39 \text{ k}\Omega, \quad R_2 = 12 \text{ k}\Omega, \\ R_C = 1 \text{ k}\Omega, \quad R_E = 470 \text{ }\Omega, \quad \beta = 100, \quad V_{BEQ} \approx 0.7 \text{ V}.$$

The divider's Thevenin equivalent is

$$V_{TH} = 12 \frac{12}{39 + 12} \approx 2.824 \text{ V}, \quad R_{TH} = 39 \parallel 12 \approx 9.176 \text{ k}\Omega.$$

The base current is

$$I_{BQ} = \frac{2.824 - 0.7}{9.176 \text{ k}\Omega + 101(470 \text{ }\Omega)} \approx 37.5 \text{ }\mu\text{A}.$$

Therefore,

$$I_{CQ} \approx 3.75 \text{ mA}, \\ I_{EQ} \approx 3.79 \text{ mA}, \\ V_{EQ} \approx (3.79 \text{ mA})(470 \text{ }\Omega) \approx 1.78 \text{ V}, \\ V_{CQ} \approx 12 - (3.75 \text{ mA})(1 \text{ k}\Omega) \approx 8.25 \text{ V}, \\ V_{CEQ} \approx 8.25 - 1.78 = 6.47 \text{ V}.$$

The base voltage is approximately $V_{BQ} = V_{EQ} + V_{BEQ} = 2.48 \text{ V}$, slightly below the unloaded divider voltage $V_{TH} = 2.824 \text{ V}$. Base current loads the divider, which is why we retained R_{TH} in Equation (5).

3.4 Why this bias is more stable

Suppose collector and emitter currents begin to increase. The emitter voltage $V_E = I_E R_E$ rises. If the divider holds base voltage reasonably steady, $V_{BE} = V_B - V_E$ then falls. The reduced V_{BE} opposes the current rise.

A useful approximation is available when the divider is stiff enough that base-current loading is small:

$$V_B \approx V_{TH}, \quad I_E \approx \frac{V_{TH} - 0.7 \text{ V}}{R_E}.$$

Current is then determined mainly by a voltage and R_E , rather than by β . Use Equation (5) when a reliable numerical answer is needed.

Bias and signal behavior are separate. A capacitor placed across R_E is open for DC, so it does not remove this DC stabilizing action. At signal frequencies where the capacitor acts as a short, it can remove much of R_E 's AC feedback.

4 Before comparing amplifier configurations

The names *common emitter*, *common base*, and *common collector* tell us which transistor terminal is shared by the input and output signal circuits. “Common” usually means at approximately *AC ground*; that terminal need not be physically connected to DC ground.

For the following midband derivations:

- DC voltage supplies are AC ground.
- Coupling and bypass capacitors are assumed to act approximately as AC shorts.
- The transistor operates near its forward-active Q-point.
- We initially neglect the Early effect: $r_o \rightarrow \infty$.

Write the effective collector resistance as

$$R'_C = R_C \parallel R_L$$

when an AC-coupled collector load R_L is present. If no such load is present, $R'_C = R_C$.

5 Common-emitter amplifier

The signal enters the base and the output is taken at the collector. The emitter is the common terminal. This configuration can provide substantial voltage gain and inverts the signal.

5.1 Gain with an AC-grounded emitter

If the emitter is AC-grounded, $v_{be} = v_b$. The small-signal collector current is

$$i_c = g_m v_{be} = g_m v_b.$$

An increase in collector current lowers collector voltage by $i_c R'_C$. Hence

$$v_o = -i_c R'_C = -g_m R'_C v_b,$$

giving

$$A_{v,CE} \equiv \frac{v_o}{v_b} \approx -g_m R'_C = -\frac{\beta R'_C}{r_\pi}. \quad (9)$$

The minus sign means the collector output is inverted relative to the base input.

5.2 Gain with an unbypassed emitter resistor

If R_E is not bypassed, base current i_b produces

$$v_{be} = i_b r_\pi, \quad v_e = i_e R_E \approx (\beta + 1) i_b R_E.$$

Thus

$$v_b = i_b [r_\pi + (\beta + 1) R_E].$$

Since $v_o = -\beta i_b R'_C$,

$$A_{v,CE} \approx -\frac{\beta R'_C}{r_\pi + (\beta + 1) R_E}. \quad (10)$$

For large β , this is approximately

$$A_{v,CE} \approx -\frac{R'_C}{r_e + R_E}.$$

The emitter resistor decreases gain because it makes the emitter follow part of the input signal, reducing the resulting v_{be} .

5.3 Input resistance

Looking into the base, the signal sees

$$R_{in,base} \approx r_\pi + (\beta + 1) R_E. \quad (11)$$

Set $R_E = 0$ in this equation when the emitter is effectively AC-grounded. If a bias divider is connected to the base, the resistance seen at the amplifier input terminal is

$$R_{in,CE} \approx (R_1 \parallel R_2) \parallel [r_\pi + (\beta + 1) R_E].$$

Thus adding an unbypassed R_E raises the resistance looking into the transistor base, although the bias divider may limit the total input resistance.

5.4 Output resistance

Set the independent input source to zero and remove R_L . With r_o neglected, the controlled source produces no additional output current when the base and emitter are AC-grounded. The output then sees R_C to AC ground:

$$R_{out,CE} \approx R_C. \quad (12)$$

For an AC-grounded emitter, including a finite r_o changes this estimate to $R_C \parallel r_o$.

6 Common-base amplifier

The signal enters at the emitter, the base is held at AC ground, and the output is taken at the collector. It provides voltage gain without inverting the input signal, but its input resistance is low.

6.1 Voltage gain

Because the base is at AC ground, an emitter input voltage v_e gives $v_{be} = v_b - v_e = -v_e$. The controlled collector current change is

$$i_c = g_m v_{be} = -g_m v_e.$$

Collector voltage changes by $v_o = -i_c R'_C$. Therefore,

$$\boxed{A_{v,\text{CB}} \equiv \frac{v_o}{v_e} \approx +g_m R'_C.} \quad (13)$$

A rising emitter voltage reduces v_{be} and collector current; the smaller drop across R'_C makes collector voltage rise. Input and output therefore move in the *same* direction.

6.2 Input resistance

Apply a small test voltage v_e at the emitter while holding the base at AC ground. The magnitude of base-emitter voltage change is v_e . The emitter current entering the transistor has two small-signal contributions:

$$i_{\text{in}} = g_m v_e + \frac{v_e}{r_\pi}.$$

Consequently,

$$\boxed{R_{\text{in,CB}} = \frac{v_e}{i_{\text{in}}} = \frac{1}{g_m + 1/r_\pi} = \frac{r_\pi}{\beta + 1} \approx \frac{1}{g_m} = r_e.} \quad (14)$$

This low resistance is a defining feature of the common-base stage. Any external resistor connected from emitter to AC ground appears in parallel with this transistor input resistance.

6.3 Output resistance

With the input signal set to zero and r_o neglected, the collector sees approximately R_C :

$$\boxed{R_{\text{out,CB}} \approx R_C.}$$

A finite r_o affects the exact output resistance, especially when R_C is large.

7 Common-collector amplifier: the emitter follower

The signal enters the base, the output is taken at the emitter, and the collector is connected to AC ground through the supply. This configuration is called an *emitter follower*: output voltage follows the input voltage without inversion.

Let R'_E be the total AC resistance from emitter to ground. For example, if there is an emitter bias resistor R_E and an AC-coupled load R_L ,

$$R'_E = R_E \parallel R_L.$$

7.1 Voltage gain

Using base current i_b ,

$$v_{be} = i_b r_\pi, \quad i_e \approx (\beta + 1)i_b, \quad v_o = v_e = (\beta + 1)i_b R'_E.$$

The base input voltage is the sum

$$v_b = v_{be} + v_e = i_b [r_\pi + (\beta + 1)R'_E].$$

Therefore,

$$A_{v,CC} \equiv \frac{v_o}{v_b} \approx \frac{(\beta + 1)R'_E}{r_\pi + (\beta + 1)R'_E}. \quad (15)$$

For large β , this becomes

$$A_{v,CC} \approx \frac{R'_E}{r_e + R'_E}.$$

The gain is positive and less than one. As base voltage rises, emitter voltage rises too; that reduces the change in v_{be} , preventing the emitter from perfectly catching up with the base.

Why use an amplifier with voltage gain below one? The emitter follower can take a signal from a source that cannot supply much current and drive a lower-resistance load. Its value is its high input resistance and relatively low output resistance, not voltage amplification.

7.2 Input resistance

The same base-current calculation gives

$$R_{in,base,CC} \approx r_\pi + (\beta + 1)R'_E. \quad (16)$$

The emitter load appears approximately $\beta + 1$ times larger when viewed from the base. Include any bias network in parallel to obtain the resistance seen at the amplifier input terminal.

7.3 Output resistance

For output resistance, remove the external load R_L and set the independent input voltage source to zero. Let R_G be the effective resistance from base to AC ground under this test. If the source has resistance R_S and the base bias network has equivalent resistance R_B , then

$$R_G = R_S \parallel R_B.$$

Looking into the emitter, the transistor presents approximately

$$\frac{r_\pi + R_G}{\beta + 1}.$$

An emitter bias resistor R_E remains connected to AC ground, so the total output resistance is

$$R_{out,CC} \approx R_E \parallel \frac{r_\pi + R_G}{\beta + 1}. \quad (17)$$

If there is no separate emitter bias resistor, omit the parallel R_E term. Notice that output resistance depends on how firmly the *base is held* when the input source is set to zero.

8 One numerical comparison

Use the voltage-divider example's approximate $I_{CQ} = 3.75$ mA, $\beta = 100$, and $V_T = 26$ mV. The corresponding small-signal parameters are

$$g_m \approx \frac{3.75 \text{ mA}}{26 \text{ mV}} = 0.144 \text{ S}, \quad r_\pi \approx \frac{100}{0.144} \approx 694 \text{ } \Omega, \quad r_e \approx 6.94 \text{ } \Omega.$$

The following are *illustrations of the configuration formulas*, not a claim that the same DC bias network can be moved unchanged between all three connections.

Configuration	Assumed AC resistance	Approximate result
CE, emitter AC-grounded	$R'_C = 1 \text{ k}\Omega$	$A_v \approx -144$
CE, R_E unby-passed	$R'_C = 1 \text{ k}\Omega$, $R_E = 470 \text{ } \Omega$	$A_v \approx -2.07$
CB	$R'_C = 1 \text{ k}\Omega$	$A_v \approx +144$; $R_{in} \approx 6.87 \text{ } \Omega$
CC	$R'_E = 470 \text{ } \Omega$	$A_v \approx +0.985$; $R_{in,base} \approx 48.2 \text{ k}\Omega$

For the unbypassed CE result, we used

$$A_v \approx -\frac{100(1000)}{694 + 101(470)} \approx -2.07.$$

For the emitter follower,

$$A_v \approx \frac{101(470)}{694 + 101(470)} \approx 0.985.$$

The large magnitudes 144 for the idealized CE and CB examples are *local small-signal gains*. Real loading, finite r_o , and the available voltage swing reduce what can be obtained in a practical circuit. A large input will cause clipping rather than an arbitrarily large output.

9 Comparison of the three configurations

Configuration	Input terminal	Output terminal	Phase	Typical resistance behavior
Common emitter	Base	Collector	Inverted	Moderate input; relatively high output
Common base	Emitter	Collector	Not inverted	Very low input; relatively high output
Common collector	Base	Emitter	Not inverted	High input; low output

Quick intuition. CE trades input-voltage changes for amplified, inverted collector voltage changes. CB accepts input current readily because the emitter input resistance is low. CC lets its emitter voltage follow the base and is useful for driving a load.

10 Source resistance, loading, and the Early effect

10.1 The source-to-output gain can differ

If a source voltage v_s has series resistance R_S and drives an amplifier with input resistance R_{in} ,

$$v_{in} = v_s \frac{R_{in}}{R_S + R_{in}}.$$

Thus, for any configuration,

$$\boxed{\frac{v_o}{v_s} = \frac{v_o}{v_{in}} \frac{R_{in}}{R_S + R_{in}}}. \quad (18)$$

This is particularly consequential for CB: its low input resistance can cause much of a voltage source's signal to fall across R_S .

10.2 What changes when r_o is finite?

The formulas above began with $r_o \rightarrow \infty$. A real transistor's collector current changes slightly with v_{ce} , represented in the hybrid- π model by r_o between collector and emitter.

For a CE stage with its emitter firmly at AC ground, a useful gain correction is

$$A_{v,CE} \approx -g_m(R_C \parallel R_L \parallel r_o),$$

and, with the external load removed,

$$R_{out,CE} \approx R_C \parallel r_o.$$

When r_o is much larger than the surrounding resistances, the simpler formulas are usually adequate. With emitter degeneration or a CB/CC connection, r_o 's effect interacts with feedback; use the complete small-signal circuit when a precise answer is required.

11 A reliable problem-solving sequence

1. **Do DC analysis first:** open the capacitors, find the base, emitter, and collector voltages, then check forward-active operation.
2. **Calculate g_m and r_π :** use the collector current at the actual Q-point.
3. **Draw the AC equivalent:** ground the DC supply for AC and determine which capacitors act as shorts at the frequency of interest.
4. **Identify the common terminal:** this tells you whether the circuit is CE, CB, or CC.
5. **Derive gain at the amplifier input terminal:** include the effective AC load.
6. **Find input and output resistances:** include bias resistors and the source termination where appropriate.
7. **Account for source loading:** use Equation (18) if gain from v_s is requested.
8. **Check the predicted swing:** use the load line and cutoff/saturation limits to ensure that a small-signal result is applicable.